



A simple division-free algorithm for computing determinants

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ABSTRACT

We present an extremely simple method for computing determinants, one that uses no division operations, exact or otherwise. The method amounts to no more than iterating a certain matrix multiplication and requires $O(nM(n))$ additions and multiplications for an $n \times n$ matrix, where $M(n)$ is the number of such operations needed for matrix multiplication. A direct combinatorial proof of correctness is given.

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1. Introduction

Let $X = (x_{ij})$ be an $n \times n$ matrix. Define $\mu(X)$ to be the matrix

$$\begin{pmatrix} -\sum_{j=2}^n x_{jj} & x_{12} & \cdots & x_{1n} \\ 0 & -\sum_{j=3}^n x_{jj} & \cdots & x_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & -\sum_{j=n+1}^n x_{jj} \end{pmatrix}$$

Thus entries of X below the diagonal are made zero, those above the diagonal are left unchanged, and each diagonal entry is replaced by the negated sum of the elements in the diagonal below it. Note that $\sum_{j=n+1}^n x_{jj} = 0$.

Define F_A by $F_A(X) = \mu(X)A$ and consider the matrix $F_A^{n-1}(A)$ in which F_A is applied $n-1$ times to A .

Theorem 1. $F_A^{n-1}(A)$ is everywhere zero except for its leading entry, which equals $|A|$ if n is odd and $-|A|$ otherwise.

Computing $\mu(X)A$ takes $O(M(n))$ additions and multiplications and, since this computation is repeated $n-1$ times, the total complexity for computing the determinant $|A|$ of an $n \times n$ matrix A is $O(nM(n))$ steps. It is difficult to imagine a simpler procedure for computing the determinant.

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2. Proof of theorem

We establish an explicit formula for the entries in the matrix $F_A^p(A)$, from which the main result will follow. To this end we employ some notation from Knuth [3]. For a fixed $n \times n$ matrix A , define $f[\alpha, \beta]$ to be the determinant of the submatrix of A formed by taking rows α and columns β , where α and β denote words of the same length over the alphabet $\{1 \dots n\}$. In particular, $f[i, j] = a_{ij}$ (roman letters denote words of length 1) and $f[\alpha, \beta] = 0$ if α or β contains a duplicated element.

Next, let $S_p(\alpha)$ denote the set of subsequences of α of length p . We will prove by induction on p that the (i, j) th entry $x_{ij}^{(p)}$ of $F_A^p(A)$ is given by

$$x_{ij}^{(p)} = (-1)^p \left(\sum f[i\alpha, j\alpha] \mid \alpha \in S_p([i+1 \dots n]) \right) \quad (1)$$

[The expression $i\alpha$ denotes the word consisting of i followed by α .] Given (1), note that $S_{n-1}([i+1 \dots n])$ is the empty set unless $i = 1$, in which case it is the singleton set $\{[2 \dots n]\}$. Hence

$$x_{11}^{(n-1)} = (-1)^{n-1} f[[1 \dots n], [1 \dots n]] = (-1)^{n-1} |A|$$

Note also that $x_{ij}^{(n-1)} = 0$ unless $i = j = 1$ because for $j \neq 1$ the word consisting of j followed by $[2 \dots n]$ contains j twice. These two facts establish the theorem.

It remains to prove (1). For the base case note that $S_0(\alpha) = \{\epsilon\}$, where ϵ denotes the empty word. Hence $x_{ij}^{(0)} = f[i, j] = a_{ij}$ as required.

For the induction step, let $d_i^{(p)}$ denote the i th diagonal element of $\mu(F_A^p(A))$, so

$$d_i^{(p)} = -\left(\sum x_{kk}^{(p)} \mid k \in \beta_i\right)$$

where $\beta_i = [i+1 .. n]$. Using (1) as induction hypothesis, we have that $d_i^{(p)}$ equals

$$(-1)^{(p+1)} \left(\sum f[k\alpha, k\alpha] \mid k \in \beta_i, \alpha \in S_p(\beta_k)\right)$$

Equivalently, since $S_{p+1}(\beta_i) = \{k\alpha \mid k \in \beta_i, \alpha \in S_p(\beta_k)\}$,

$$d_i^{(p)} = (-1)^{(p+1)} \left(\sum f[\alpha, \alpha] \mid \alpha \in S_{p+1}(\beta_i)\right) \quad (2)$$

Now, the (i, j) th entry of $\mu(F_A^p(A))A$ is given by

$$d_i^{(p)} a_{ij} + \left(\sum x_{ik}^{(p)} a_{kj} \mid k \in \beta_i\right)$$

Using (2) and the induction hypothesis (1) again, we obtain that the (i, j) th entry is $(-1)^{(p+1)}$ times

$$\begin{aligned} &\left(\sum f[\alpha, \alpha] a_{ij} \mid \alpha \in S_{p+1}(\beta_i)\right) \\ &- \left(\sum f[i\alpha, k\alpha] a_{kj} \mid \alpha \in S_p(\beta_i), k \in \beta_i\right) \end{aligned} \quad (3)$$

Our task is to show that (3) equals

$$(-1)^{(p+1)} \left(\sum f[i\alpha, j\alpha] \mid \alpha \in S_{p+1}(\beta_i)\right) \quad (4)$$

Expanding the determinant $f[i\alpha, j\alpha]$ by its first column, we have

$$\begin{aligned} f[i\alpha, j\alpha] &= a_{ij} f[\alpha, \alpha] - \left(\sum a_{kj} f[i(\alpha \setminus h), h(\alpha \setminus k)] \mid k \in \alpha\right) \end{aligned}$$

where $\alpha \setminus \beta$ is what remains when the symbols in β are removed from α . Hence (4) equals $(-1)^{(p+1)}$ times

$$\begin{aligned} &\left(\sum a_{ij} f[\alpha, \alpha] \mid \alpha \in S_{p+1}(\beta_i)\right) \\ &- \left(\sum a_{kj} f[i(\alpha \setminus k), h(\alpha \setminus k)] \mid \alpha \in S_{p+1}(\beta_i), k \in \alpha\right) \end{aligned} \quad (5)$$

Comparing (3) and (5), it is therefore sufficient to show

$$\begin{aligned} &\left(\sum f[i\alpha, k\alpha] \mid \alpha \in S_p(\beta), k \in \beta\right) \\ &= \left(\sum f[i(\alpha \setminus k), k(\alpha \setminus k)] \mid \alpha \in S_{p+1}(\beta), k \in \alpha\right) \end{aligned}$$

for any word β not containing duplicates. Indeed, since $f[i\alpha, k\alpha] = 0$ if α contains k , it is sufficient to show that

$$\begin{aligned} &\{(i\alpha, k\alpha) \mid \alpha \in S_p(\beta), k \in \beta \setminus \alpha\} \\ &= \{(i(\alpha' \setminus k), k(\alpha' \setminus k)) \mid \alpha' \in S_{p+1}(\beta), k \in \alpha'\} \end{aligned}$$

Take $\alpha' \in S_{p+1}(\beta)$ and $k \in \alpha'$. Then $\alpha' \setminus k \in S_p(\beta)$ and $k \in \beta \setminus \alpha$. Conversely, take $\alpha \in S_p(\beta)$ and $k \in \beta \setminus \alpha$. Then k can be inserted in a unique position in α to give a word $\alpha' \in S_{p+1}(\beta)$. Furthermore, $\alpha = \alpha' \setminus k$.

3. Discussion

Although it doesn't reduce the asymptotic complexity, the algorithm can be speeded up by taking advantage of the fact that in computing $\mu(X)A$ we need only the upper-triangular result. Furthermore, the number of rows that need be considered reduces by one at each iteration. In my book [2] the algorithm is implemented in the purely functional language Haskell, and compared to two other algorithms, Gaussian elimination with rational arithmetic, and Bareiss's method [1] that uses exact integer division. Both these algorithms take $\Theta(n^3)$ arithmetic operations, though Gaussian elimination requires expensive rational division and in Bareiss's method the integers can get big quite quickly. Unlike these algorithms our method does not require pivoting, though it does require $\Theta(n^4)$ arithmetic operations if naive matrix multiplication is used. We used the Haskell types *Integer* and *Ratio Integer* that provide exact arbitrary precision integers and rational arithmetic, and implemented matrices as lists of lists. For 100×100 matrices it turns out that our method is slightly slower than the Bareiss algorithm, and about the same speed as Gaussian elimination. As far as I am aware, the goal of a $\Theta(n^3)$ division-free determinant calculation remains elusive.

As to its history, our algorithm was extracted from a division-free algorithm of Mahajan and Vinay [4] based on the idea of *clow* sequences (see also the survey [5] for additional information about clow sequences and division-free algorithms). The word *clow* is an acronym for "closed ordered walk", and a clow sequence is a generalisation of the cycle decomposition of a permutation in which each cycle can contain repetitions of intermediate elements, hence a closed walk. Mahajan and Vinay showed that all the signed matrix terms for clow sequences that do not correspond to permutations cancel each other out, leaving just those terms $a_{1\pi(1)} a_{2\pi(2)} \dots a_{n\pi(n)}$ that do correspond to permutations. But their proof is not trivial. The signed matrix term corresponding to a clow sequence can be expressed as a path in a layered directed acyclic graph to one of two endpoints, and the computation of the sum of such factors computed as a path problem. By recasting the associated recursive definition directly back into matrix operations, the algorithm described above was discovered. There was no proof in [2] that the method was correct, and this has been repaired in the current paper with a combinatorial proof that does not depend on any of the numerous identities about determinants that have been discovered in the last 250 years.

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References

- [1] E.H. Bareiss, Sylvester's identity and multi-step integer preserving Gaussian elimination, *Math. Comp.* 22 (103) (1968) 565–578.
- [2] Richard Bird, *Pearls of Functional Algorithm Design*, Cambridge University Press, UK, 2010.

- [3] D.E. Knuth, Overlapping Pfaffians, *Electron. J. Combin.* 3 (2) (1996), Article R5.
- [4] M. Mahajan, V. Vinay, Determinant: Combinatorics, algorithms and complexity, *Chic. J. Theoret. Comput. Sci.* 5 (1997).
- [5] G. Rote, Division-free algorithms for the determinant and the Pfaffian: Algebraic and combinatorial approaches, in: H. Alt (Ed.), *Computational Discrete Mathematics*, in: LNCS, vol. 2122, Springer-Verlag, 2001, pp. 119–135.